

COSC264 · INTRODUCTION TO COMPUTER NETWORKS AND THE INTERNET · TERM 4 · MODULE 6

# Application Protocols: the Web, Email and DNS

1 · Network applications

2 · The Web and HTTP

3 · Email

4 · DNS

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OUTLINE

# Module 6 — roadmap

**General concepts about network applications**

**One network application: the Web**

**HTTP: the protocol that supports the Web**

**Email**

**DNS: the Domain Name System**

MODULE 6

# Network Applications

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What an application needs from the layers beneath it

1 • Network applications

2 • The Web and HTTP

3 • Email

4 • DNS

# What is a network application?

**Programs that run on different end systems** and communicate with each other over the network.

## Web browsing

Google Chrome, Mozilla Firefox,  
Microsoft Edge

## Email

Gmail, Outlook

## Instant messaging

Slack, WhatsApp, Facebook  
Messenger

## Streaming services

YouTube, Netflix

## Social media

Instagram, TikTok

## AI assistants

Claude, ChatGPT, Gemini

# Application-layer protocols

An **application-layer protocol** is one piece of a network application — but a big piece.

It defines how the application's processes, running on different end systems, **pass messages to each other**: which messages exist, their format, and when and how to respond.

This is what lets programs on different machines work together — e.g. any web browser can fetch a page from any web server, whoever wrote each program.

| Network application | Application-layer protocol |
|---------------------|----------------------------|
| Web application     | HTTP                       |
| Email               | SMTP                       |

## What applications need from the transport layer

- Application-layer protocols are controlled by the **application developer**; transport-layer protocols, however, are built into the **OS**.
- As a developer you can only choose between the protocols the OS provides — so the choice matters.

| Property               | Example applications                                    | Transport protocol                                      |
|------------------------|---------------------------------------------------------|---------------------------------------------------------|
| Reliable data transfer | Email, IM, file transfer, financial apps                | <b>TCP</b> preferred<br>UDP unsuitable                  |
| Low delay              | Internet telephony, teleconferencing, multiplayer games | <b>UDP</b> preferred<br>TCP retransmissions add latency |

MODULE 6

# The Web and HTTP

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The application-layer protocol behind every page you load

1 · Network applications

2 · **The Web and HTTP**

3 · Email

4 · DNS

# The World Wide Web and HTTP

## THE WEB

The **World Wide Web** (aka Web) is a global collection of documents and other resources to be shared over the Internet — like a large file storage system. Anyone can request a file and get it back.

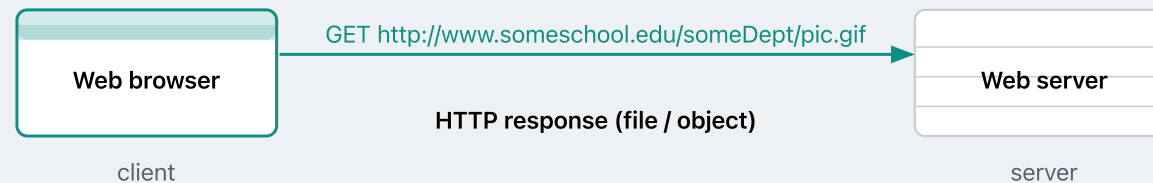
Each object on the Web (e.g., an image, an HTML file) is addressable by a **URL**:

```
http://www.someschool.edu/someDept/pic.gif
```

└─ host name ─┘ └─ path name ─┘

## HTTP

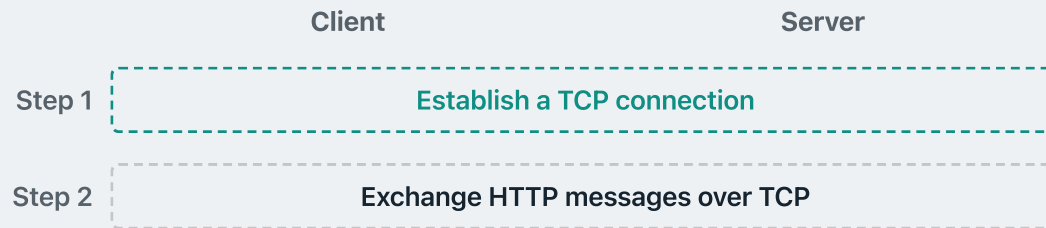
The **Hypertext Transfer Protocol** is the application-layer protocol that supports the Web — it defines how clients request objects by URL and how servers respond.



HTTP

# How HTTP works

HTTP uses **TCP** as its transport protocol — so before any HTTP message is sent, a TCP connection must be established first. (HTTP/3 uses QUIC instead — an exception.)



## STATE LESS

HTTP is **stateless** — the server maintains no information about past client requests. Each request is handled independently.

# HTTP connections

## Non-persistent HTTP

- At most **one object** is sent over a TCP connection — connection is then closed
- HTTP/1.0 uses non-persistent HTTP

## Persistent HTTP

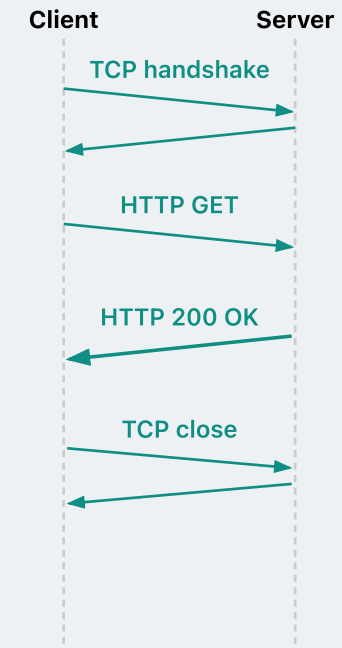
- **Multiple objects** can be sent over a single TCP connection
- HTTP/1.1, HTTP/2, and HTTP/3 use persistent HTTP

| Protocol | Published | Connection type   | Status   |
|----------|-----------|-------------------|----------|
| HTTP/1.0 | 1996      | Non-persistent    | Legacy   |
| HTTP/1.1 | 1997      | Persistent        | Standard |
| HTTP/2   | 2015      | Persistent        | Standard |
| HTTP/3   | 2022      | Persistent (QUIC) | Standard |

# Non-persistent HTTP in action

Suppose a client wants to retrieve a file from: `http://www.someSchool.edu/someDepartment/home.index`

1. Client establishes a **TCP connection** with the server
2. Client sends an **HTTP GET** request — path: `/someDepartment/home.index`
3. Server retrieves the HTML file, wraps it in an **HTTP response**, and sends it back
4. Server signals TCP to **close the connection**



## Response time: non-persistent HTTP

**RTT**

**Round Trip Time** — time for a small packet to travel from client to server and back

**RESPONSE  
TIME**

Time to retrieve a single object using HTTP

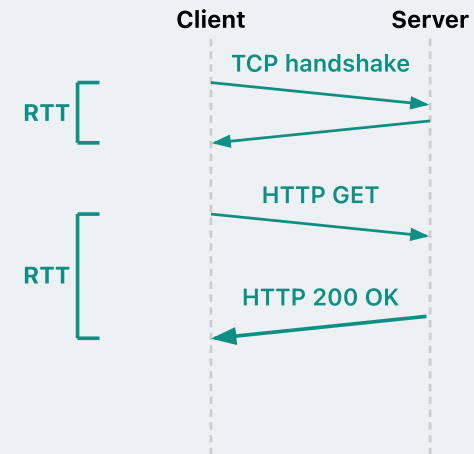
**Q:** What is the response time for non-persistent HTTP?

**A:** 2 RTTs + file transmission time

**First RTT:** the TCP handshake — client sends SYN, server replies SYN-ACK

**Second RTT:** client sends HTTP GET, server sends back the response

With many referenced objects, this cost multiplies — a significant burden on the server



## HTTP

# Persistent HTTP in action

Client and server establish a TCP connection and exchange HTTP messages

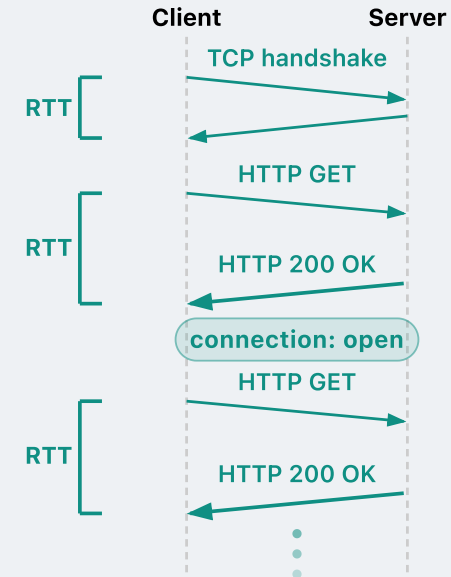
Server **keeps the TCP connection open** after sending the first HTTP response

Subsequent HTTP messages between the same client and server use the **same TCP connection** — no new handshake needed

### Response time on one persistent connection:

1 RTT for initial TCP handshake + 1 RTT per HTTP request

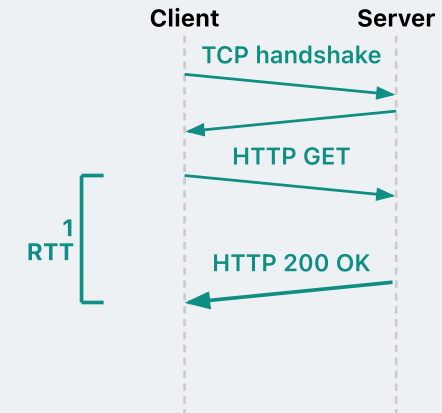
Amortised: **~1 RTT per object** for many requests



# Improving HTTP efficiency further with pipelining

**WITHOUT  
PIPELINING**

Client issues a new request **only after** receiving the previous response —  
**1 RTT per file**



# Improving HTTP efficiency further with pipelining

## WITHOUT PIPELINING

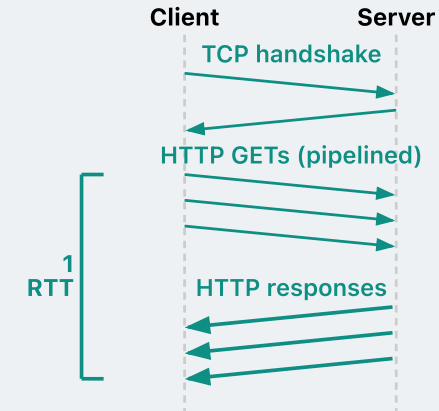
Client issues a new request **only after** receiving the previous response — **1 RTT per file**

## WITH PIPELINING

Client sends requests **as soon as it needs to retrieve an object**, without waiting for responses — e.g. in this example we pipeline 3 HTTP requests all together

**Benefit:** pipelining 3 requests → response time per file  $\approx 1 \text{ RTT} / 3$

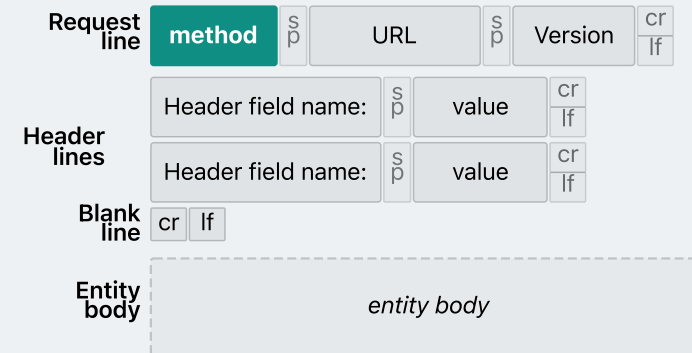
- Don't confuse this with TCP-level pipelining
- HTTP pipelining operates at the **application layer** — it pipelines **HTTP messages**
- In practice, browsers leave HTTP/1.1 pipelining off (one slow response blocks those behind it); HTTP/2 multiplexing replaced it



# HTTP request message format

Method — what action to perform

| Method        | Action                          |
|---------------|---------------------------------|
| <b>GET</b>    | Retrieve a resource             |
| HEAD          | Retrieve headers only (no body) |
| <b>POST</b>   | Submit data to a resource       |
| PUT           | Store / replace a resource      |
| <b>DELETE</b> | Remove a resource               |



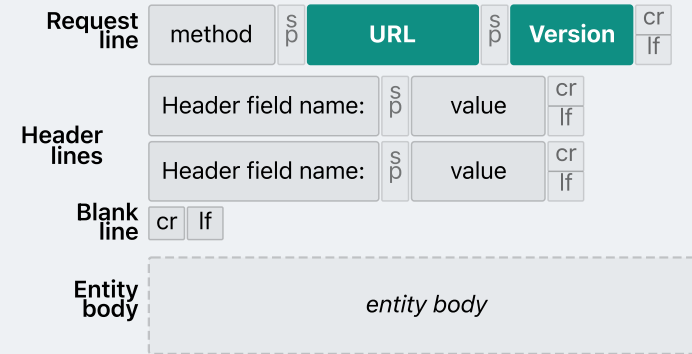
HTTP

# HTTP request message format

**URL:** path to the requested resource — e.g. /someDept/pic.gif

**HTTP Version:** identifies the protocol in use — HTTP/1.0 or

HTTP/1.1



# HTTP request message format

## Header lines

- Zero or more `field-name: value` pairs, one per line
- Common examples:

`Host: www.example.com`

Mandatory in HTTP/1.1 — identifies the target server

`Content-Type: text/html; charset=UTF-8`

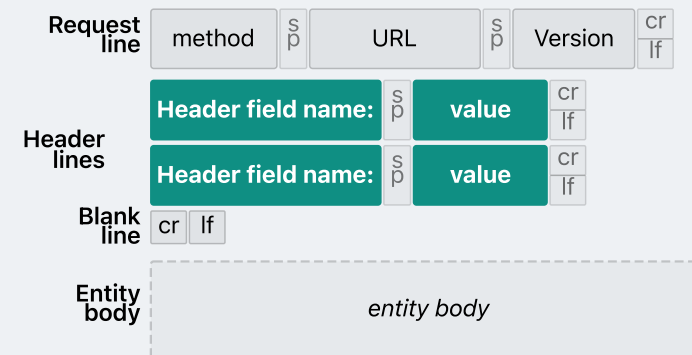
Format of the message body

`Connection: close`

Close the TCP connection after this response (non-persistent)

`Accept-Language: da, en-gb;q=0.8, en;q=0.7`

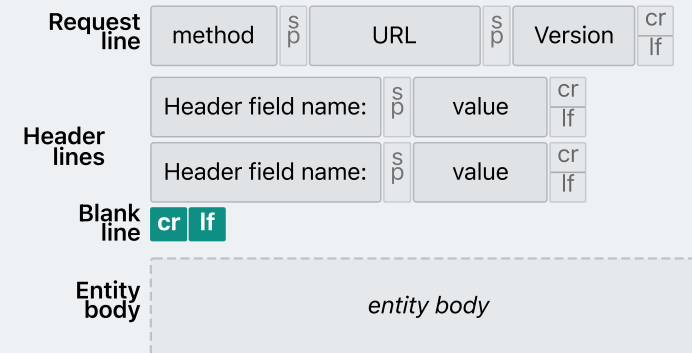
Language preference with quality weights (default  $q=1$ ): prefer Danish → British English → any English



HTTP

# HTTP request message format

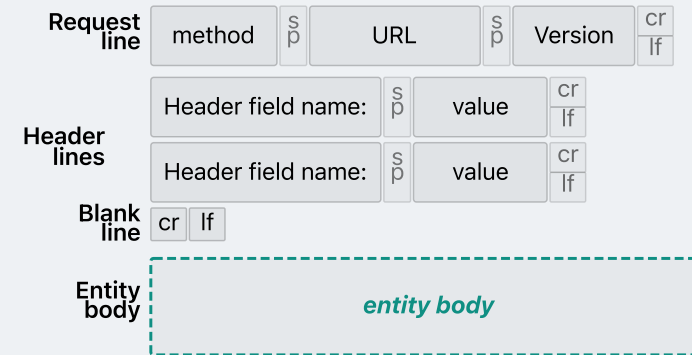
**Blank line** (a lone CRLF): marks the end of the header section



HTTP

# HTTP request message format

**Entity body:** empty for GET and HEAD; carries the payload for POST and PUT (e.g. form data or JSON)



HTTP

## HTTP request: an example

```
GET /somedir/page.html HTTP/1.1
```

```
Host: www.someschool.edu
```

```
User-agent: Mozilla/4.0
```

```
Connection: close
```

```
Accept-language: fr
```

```
-----  
(empty entity body)
```

**Method:** GET

**URL:** /somedir/page.html

**HTTP version:** HTTP/1.1

**Header field:** Host: www.someschool.edu

**Header field:** User-agent: Mozilla/4.0

**Header field:** Connection: close

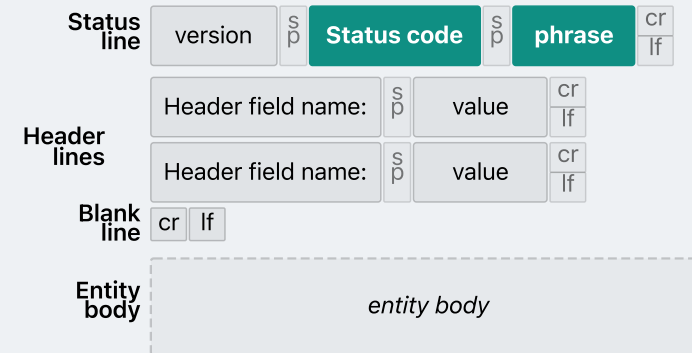
**Header field:** Accept-language: fr

**Body:** (empty)

# HTTP response message format

## Status code & phrase

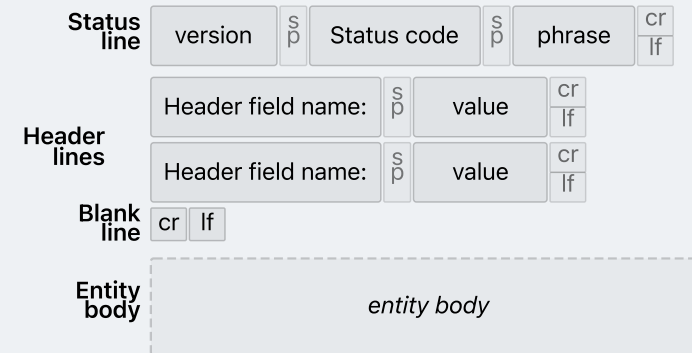
- **Status code:** 3-digit number indicating the result of the request
- **Phrase:** human-readable text accompanying the code — e.g. OK, Not Found



# HTTP response message format

## Common examples

| Code | Phrase                     |
|------|----------------------------|
| 200  | OK                         |
| 301  | Moved Permanently          |
| 400  | Bad Request                |
| 404  | Not Found                  |
| 505  | HTTP Version Not Supported |



# HTTP response: an example

**HTTP/1.1 200 OK**

Connection: close

Date: Thu, 06 Aug 1998 12:00:15 GMT

Server: Apache/1.3.0 (Unix)

Last-Modified: Mon, 22 Jun 1998 ...

Content-Length: 6821

Content-Type: text/html

-----  
<!DOCTYPE html>

<html><head><title>Example</title></head>

<body><h1>Hello!</h1>

<script>console.log("hi");</script>

</body></html>

**Status line:** HTTP/1.1 200 OK

**Body:** the requested HTML file (with embedded JS)

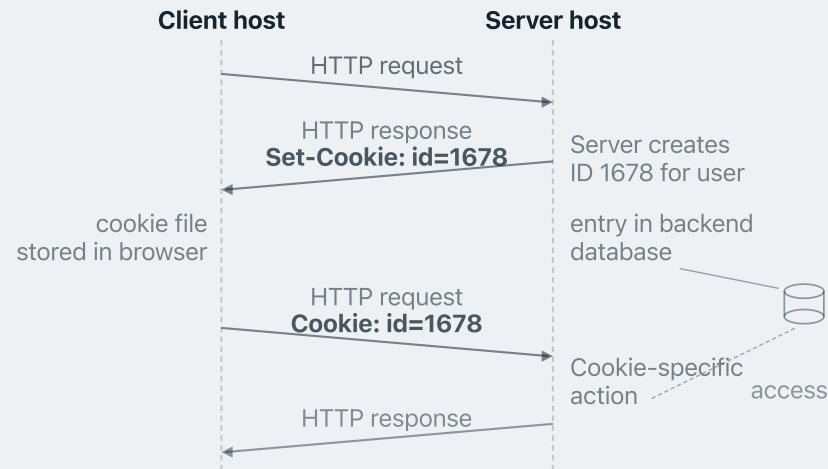
# Cookies

HTTP is **stateless** — every request is independent; the server retains no memory of past requests. But many applications *need* state. How does an online shopping site remember your cart, or recognise you across visits?

The answer: **cookies** — here's how it works.

1. A **Set-Cookie** header line in the **HTTP response** — the server sends the client a unique ID on first contact
2. A **Cookie** header line in subsequent **HTTP requests** — the client echoes the ID back on every future request
3. A **cookie file** stored on the client, managed by the browser — persists across sessions
4. A **back-end database** on the server, keyed by the cookie ID — stores per-user state

# Cookies: an example [RFC 6265]



The client sends an HTTP request to the host

**Set-Cookie: id=1678** — Server assigns a unique ID and sends it to the client

The cookie ID is stored as an entry in the back-end database, and in the client's browser

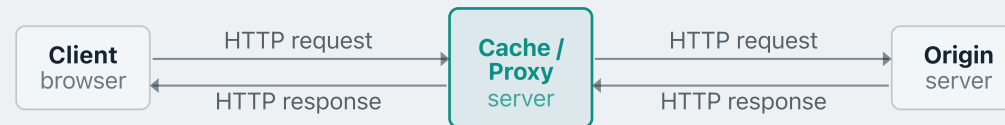
**Cookie: id=1678** — Client attaches the ID on every future request to the same server

Server uses the ID to look up the user's state in its database

Server can then generate a **tailored response** — e.g. a personalised page, a pre-filled shopping cart — for the client's request

# Web caching

A **web cache** (also called a **proxy server**) is a network entity that satisfies HTTP requests on behalf of an origin server — storing copies of recently requested objects locally.



## Acts as a server

Receives HTTP requests from clients and returns responses — just like an origin server would

## Acts as a client

When the object is not cached, it issues its own HTTP request to the origin server to fetch it

HTTP

# Why caching?

## **Faster response time**

Cached objects are served from a nearby proxy — no round trip across the internet to the origin server. Substantially lower latency for the client.

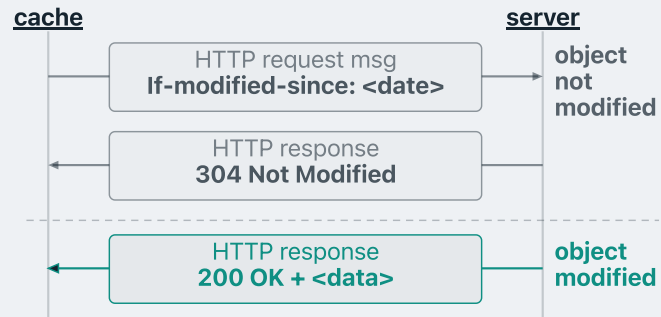
## **Reduced access-link traffic**

Requests satisfied by the cache never travel over the institution's uplink to the internet — less congestion on that bottleneck link.

## **Lower access-link costs**

Fewer objects are fetched from origin servers, so less traffic comes in over the institution's access link — it can avoid or delay a costly upgrade to that link.

# Conditional GET



**Goal:** the proxy server/cache wants to avoid re-fetching an object if its cached copy is still up-to-date

The proxy sends the following HTTP GET request:

```
GET /index.html HTTP/1.1
```

```
Host: www.example.com
```

```
If-modified-since: Mon, 10 Aug 2026 09:00:00  
GMT
```

If the object is **unchanged**, server replies:

```
304 Not Modified
```

No object in the body — cache serves its stored copy directly

Object **changed**: server replies 200 OK with the updated object in the body

MODULE 6

# Email

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Electronic Mail

1 · Network applications

2 · The Web and HTTP

**3 · Email**

4 · DNS

## OUTLINE

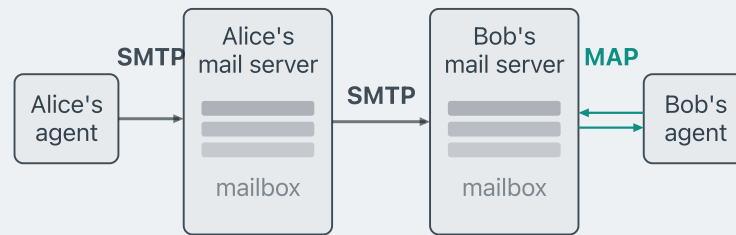
# Email — roadmap

### Electronic Mail (Email)

- *Mail transfer protocol:* **SMTP**
- *Mail access protocols:* **POP3, IMAP**
- *Message format:* **MIME**

## EMAIL

# An overview of the email system



**User agent (UA)** — the app used to read, compose and send email (e.g. Outlook, Gmail, Apple Mail)

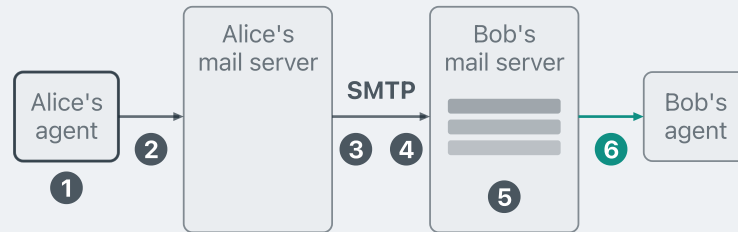
**Mail server** — a dedicated host that handles email on behalf of its users. It stores a **mailbox** per recipient for incoming mails and a **send queue** of outgoing mails

**SMTP** — Simple Mail Transfer Protocol; used to send mail, from the sender's UA to its mail server and between mail servers across the internet

**Mail access protocol (MAP)** — used by UAs to retrieve messages from their mail server; POP3 or IMAP

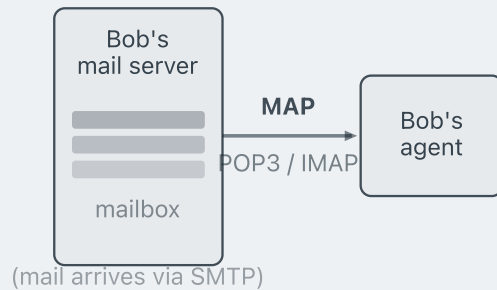
## SMTP in action

SMTP moves messages between mail servers. It consists of a **client side** and a **server side**.  
Every mail server runs both an **SMTP client** (when sending) and an **SMTP server** (when receiving).



- ① Alice uses her UA to **compose** the message
- ② Alice's UA **sends** the message to her mail server (also via SMTP)
- ③ The **SMTP client** on Alice's server opens a TCP connection to Bob's mail server
- ④ Alice's message is **transferred over the TCP connection**
- ⑤ The **SMTP server** on Bob's server receives the message and places it in Bob's **mailbox**
- ⑥ Bob invokes his UA to **read** the message

## Retrieving email from the server



**Mail access protocols** retrieve email from the mail server to the recipient's user agent

**POP3** — Post Office Protocol version 3 [RFC 1939]  
Downloads messages to the local device. Simple and stateless between sessions

**IMAP** — Internet Message Access Protocol version 4 [RFC 9051]  
Keeps messages on the server; supports folders and sync across multiple devices

# Accessing email via a browser

**Web-based email is accessed through a web browser**

Uses **HTTP** for communication between the user's browser and the mail server

**Web-based email services**

Examples: **Gmail**, **Outlook** (web)

IMAP

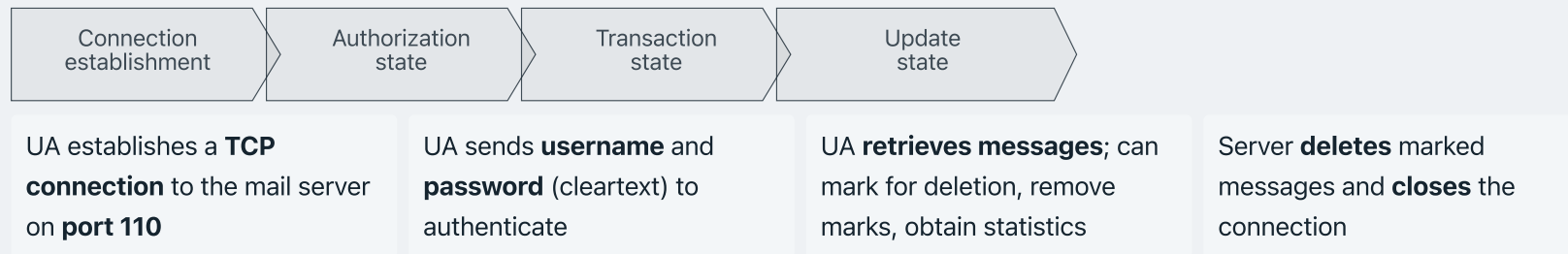
# Internet Message Access Protocol

IMAP **keeps messages on the server** and replicates copies to the client

Users can **organise messages in folders** — folder structure is stored on the server, accessible from any device

IMAP maintains **user state information** across sessions (e.g. folder names, message-to-folder mappings)

## How POP3 works



### Notable limitations:

- Does not synchronise emails across multiple devices
- Does not support server-side folder management — users cannot organise emails into folders on the server

# Structure of an email message

An email message is **ASCII text** — **header fields** followed (optionally) by a **body**

```
Date: Mon, 10 Aug 2026 09:00:00
GMT
From: alice@crepes.fr
To: bob@hamburger.edu
Subject: Picture of yummy crepe.
Cc: charlie@example.com
```

Header lines

---

```
Hello Bob, here is the picture
you asked for. Enjoy!
```

Body

## MIME

# Multipurpose Internet Mail Extensions

**Problem:** standard email is ASCII-only — it cannot natively carry images, attachments, or non-ASCII characters (e.g. Chinese, Arabic)

MIME extends the message format with additional header fields that describe and encode non-ASCII content — telling the receiver how to decode the body

e.g. **base64** encoding converts binary data into printable ASCII text, which can be used for image attachments

From: alice@crepes.fr

To: bob@hamburger.edu

Subject: Picture of yummy crepe.

**MIME-Version: 1.0**

**Content-Transfer-Encoding: base64**

**Content-Type: image/jpeg**

---

(base64 encoded data...

...base64 encoded data)

MODULE 6

# DNS

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Domain Name System

1 · Network applications

2 · The Web and HTTP

3 · Email

4 · **DNS**

# Domain Name System — overview

- As we have seen, network routing entirely relies on **IP addresses** — numerical identifiers
- However, when you browse the web, you typically type a **human-friendly hostname** such as `canterbury.ac.nz`
- How does the browser know how to translate between a hostname and an IP address?



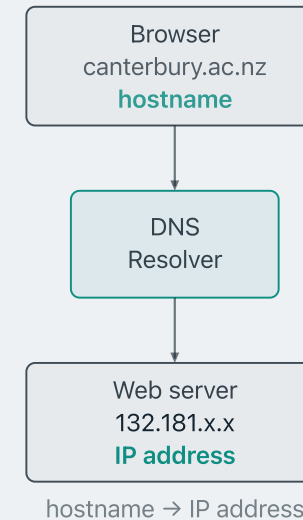
# Domain Name System — overview

- As we have seen, network routing entirely relies on **IP addresses** — numerical identifiers
- However, when you browse the web, you typically type a **human-friendly hostname** such as `canterbury.ac.nz`
- How does the browser know how to translate between a hostname and an IP address?

**DNS** is that translation mechanism — it maps human-readable **hostnames** to numerical **IP addresses**

DNS implements a **distributed database** — a global hostname-to-IP mapping distributed across a hierarchy of DNS servers [RFC 1034, 1035]

DNS is an **application-layer protocol** running mainly over **UDP on port 53** (TCP port 53 for large responses and zone transfers)



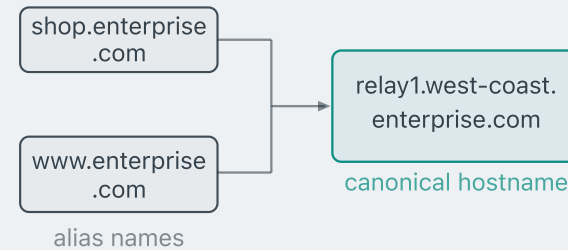
## Additional services provided by DNS

**Hostname-to-IP translation** — the primary service, already seen; commonly used by HTTP, SMTP, and other application-layer protocols

**Host aliasing** — a host with a complex **canonical hostname** can be reached via simpler **alias names**

e.g. `www.enterprise.com` and `shop.enterprise.com` both resolve to the canonical `relay1.west-coast.enterprise.com`

**Mail server aliasing** — the same idea for email; a friendly address like `bob@enterprise.com` can map (via an MX record) to the canonical mail server hostname `relay1.west-coast.enterprise.com`

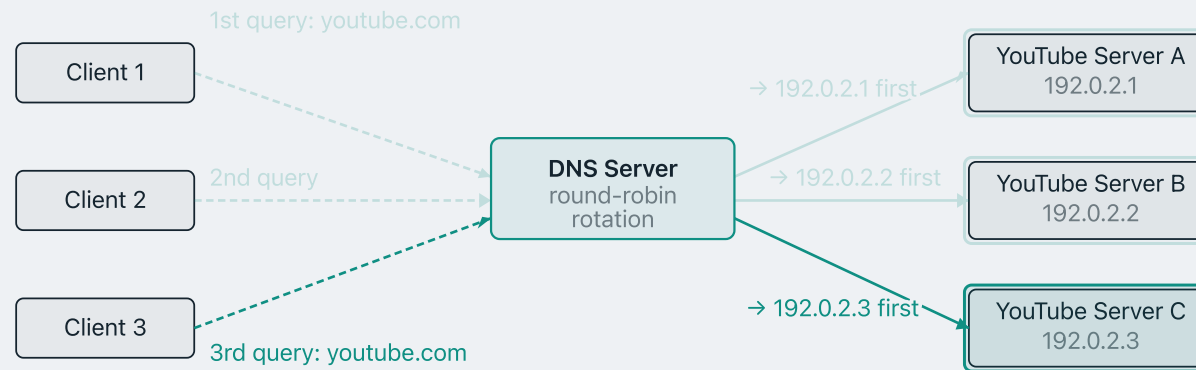


# Additional services by DNS: load distribution

Many popular web services — e.g. YouTube, Google, Netflix — are hit by **millions of users** simultaneously. A **single server** cannot handle that volume.

The service is therefore distributed across **many server instances**, each with a **distinct IP address**, all sharing the **same hostname** (e.g. `youtube.com`).

How are users assigned to these servers? **DNS load balancing** — DNS stores all the IPs and **rotates their order** on each response, so consecutive clients get a different IP listed first.



# Why not a centralised design?

A centralised design would use a **single server** that hosts the translation database — every DNS query goes to that server.

## Single point of failure

If the one server crashes, DNS lookup fails for the entire Internet

## Traffic volume

Billions of DNS queries per day — no single machine can handle that load

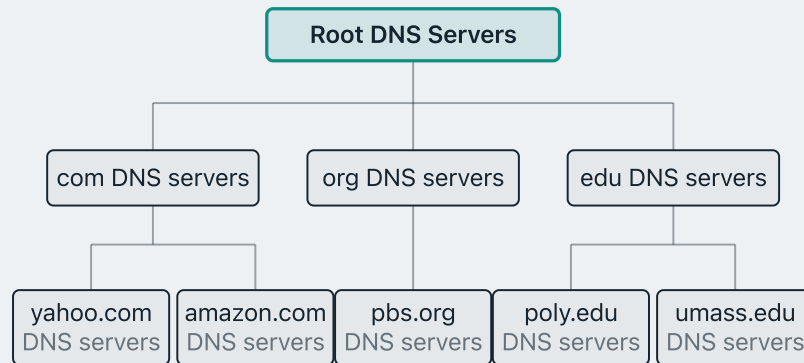
## Distant centralised database

All queries must travel to one location — users far away face high latency on every lookup

## Maintenance

Billions of hostname records change constantly — impossible to keep one database current

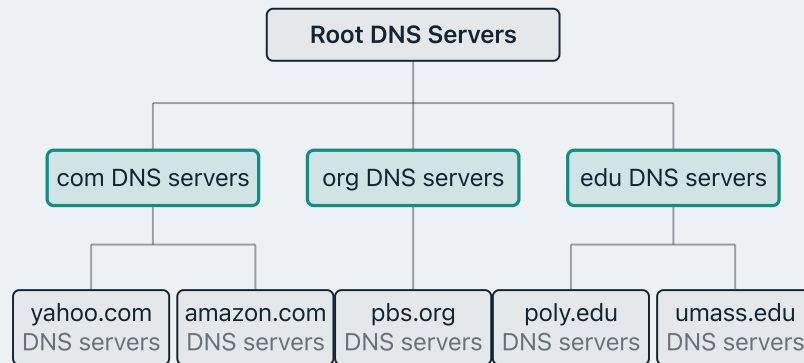
# A hierarchical distributed database



## Root DNS Servers

13 root server identities (a–m.root-servers.net) — each replicated via anycast, giving **over 1000 physical instances** worldwide

# A hierarchical distributed database

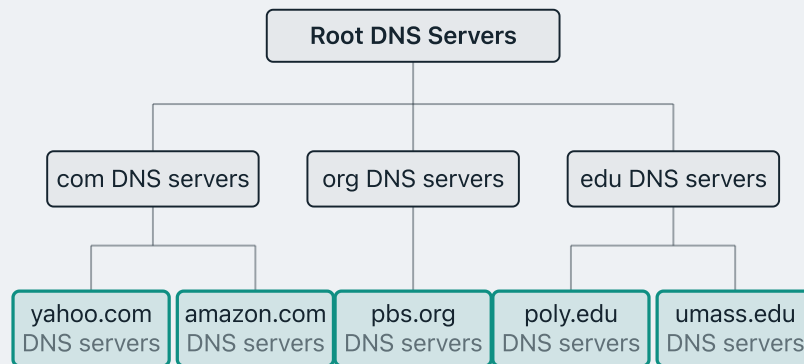


## TLD DNS Servers

Handle **.com**, **.org**, **.edu** and country-code domains (.nz, .uk). Maintained by dedicated registries — e.g.

**Verisign** (.com), **Public Interest Registry** (.org), **EDUCAUSE** (.edu), **InternetNZ** (.nz).

# A hierarchical distributed database



## Authoritative DNS Servers

Each organisation's own DNS server — the **definitive source** for its hostname-to-IP mappings.  
e.g. Canterbury's authoritative server holds the IP for `www.canterbury.ac.nz`

# Local DNS server and DNS caching

Every ISP or institution operates a **local DNS server** (also called a *default name server*). It is **not** part of the root / TLD / authoritative hierarchy — it sits **separately** from that.

When you join the network, your device is automatically configured to use one — e.g. the University of Canterbury runs its own local DNS server for all campus users.

**DNS Hierarchy**  
Root · TLD · Authoritative  
(separate from local DNS)

**Local DNS server**  
e.g. dns.canterbury.ac.nz  
(cache)

Your computer

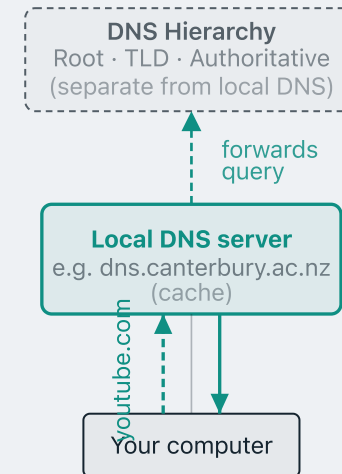
# Local DNS server and DNS caching

When your machine makes a DNS lookup, the query goes **first to the local DNS server**.

**If the answer is cached:** the local DNS server replies **immediately** — no hierarchy lookup needed.

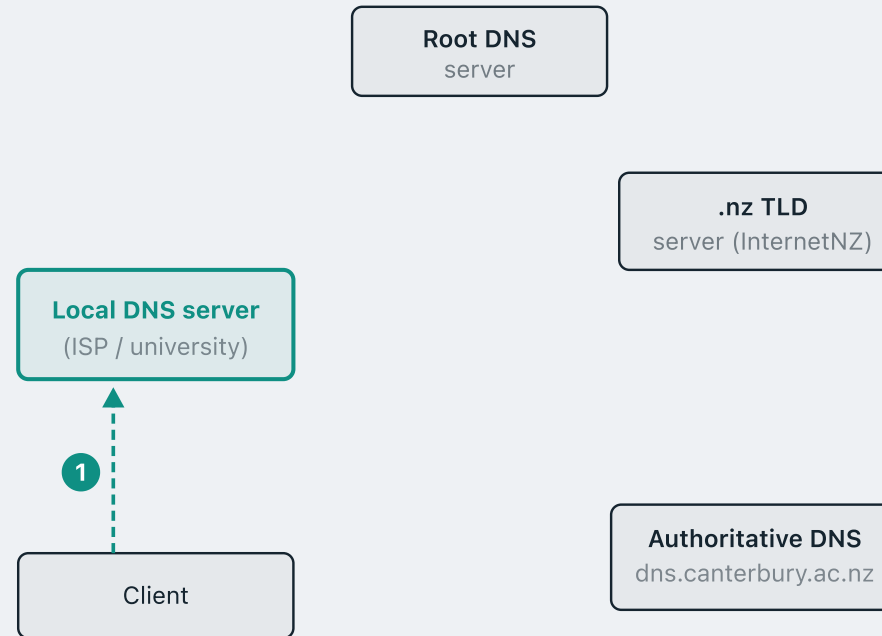
**If not cached:** the local server **forwards the query into the hierarchy**, relays the reply back to you, and **caches the hostname-to-IP mapping** in local memory for future queries.

Cache entries have a **TTL (time-to-live)** and expire after some time (typically hours). TLD server addresses are routinely cached in local DNS servers — so **root servers are bypassed for most everyday queries**.



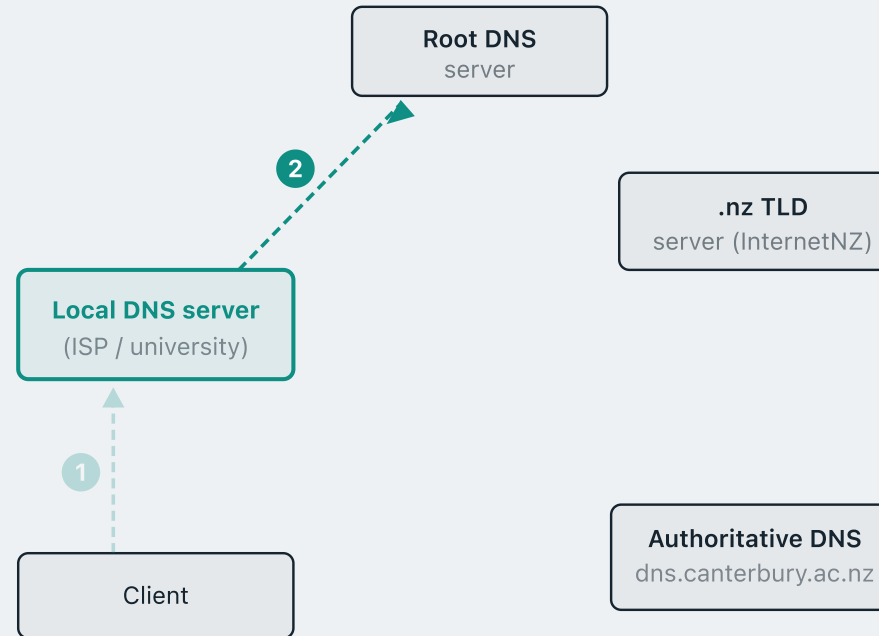
# Resolving a hostname: the full journey

**Step 1.** A client wants to resolve **csse.canterbury.ac.nz**. It sends a DNS query to its **local DNS server**.



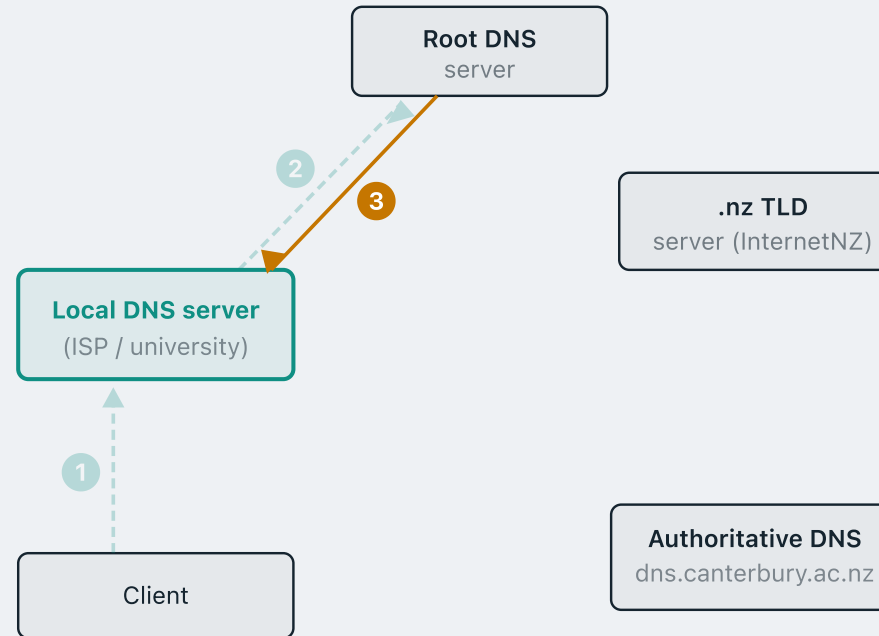
# Resolving a hostname: the full journey

**Step 2.** The local DNS doesn't have the answer cached. It contacts a **root DNS server**.



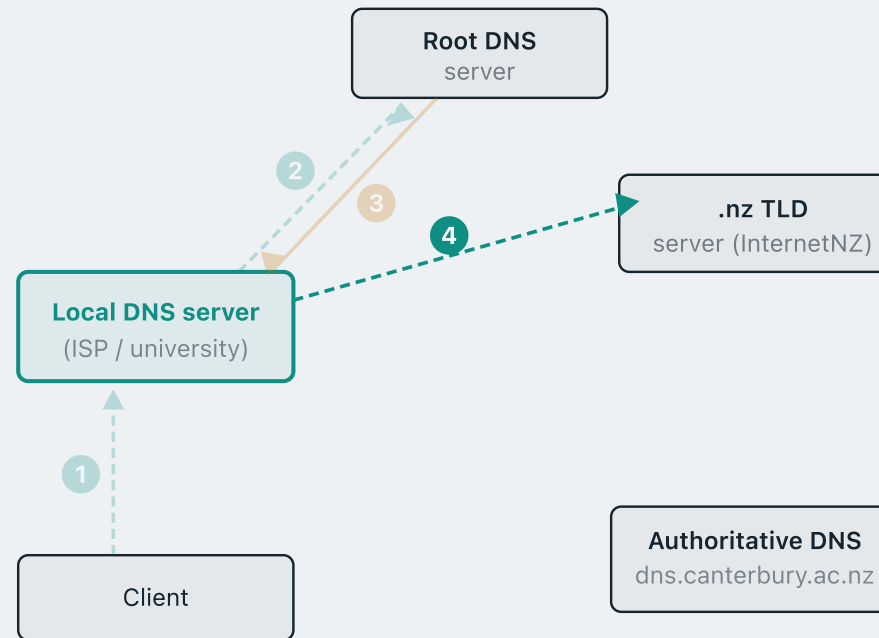
# Resolving a hostname: the full journey

**Step 3.** The root server returns a **referral**: "I don't know — try a **.nz TLD** server" and gives its address.



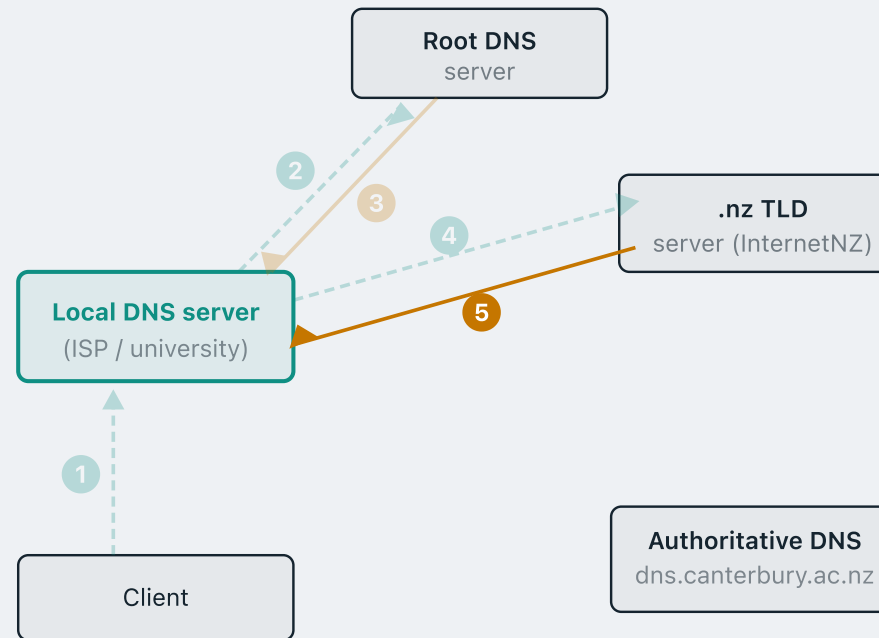
# Resolving a hostname: the full journey

**Step 4.** The local DNS follows the referral and contacts the **.nz TLD server**.



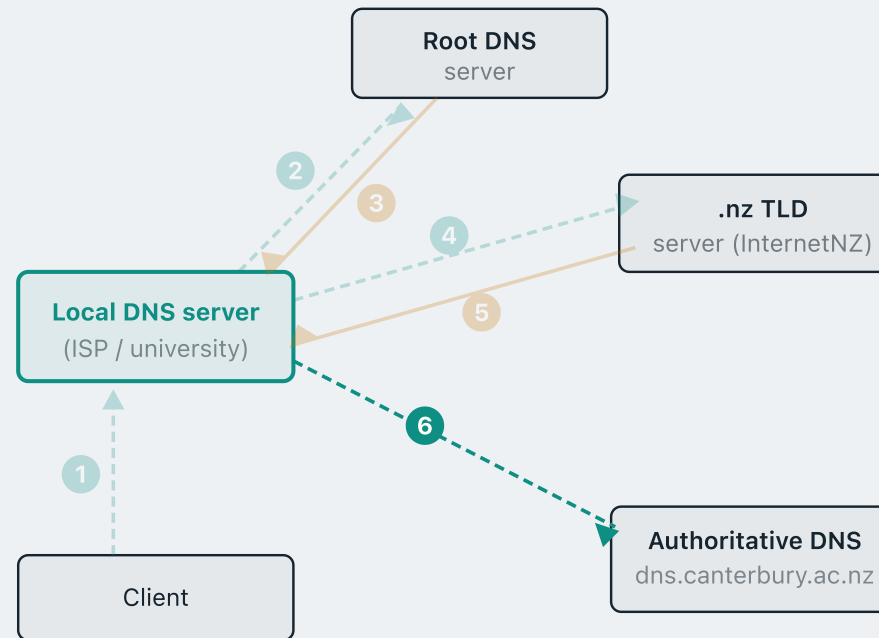
# Resolving a hostname: the full journey

**Step 5.** The .nz TLD returns another **referral**:  
"Try **dns.canterbury.ac.nz** — the  
authoritative server for canterbury.ac.nz."



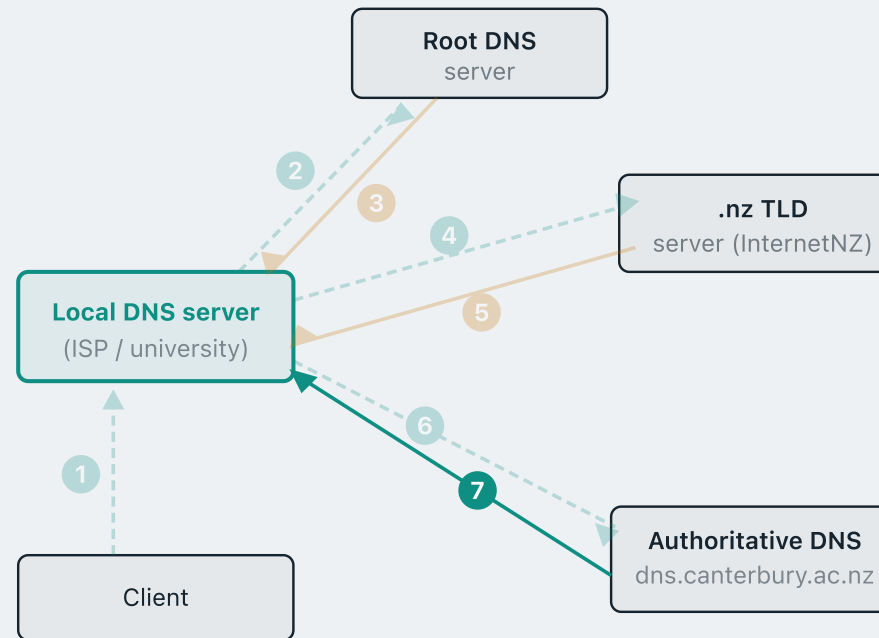
# Resolving a hostname: the full journey

**Step 6.** The local DNS contacts **dns.canterbury.ac.nz** — the authoritative name server for the canterbury.ac.nz zone.



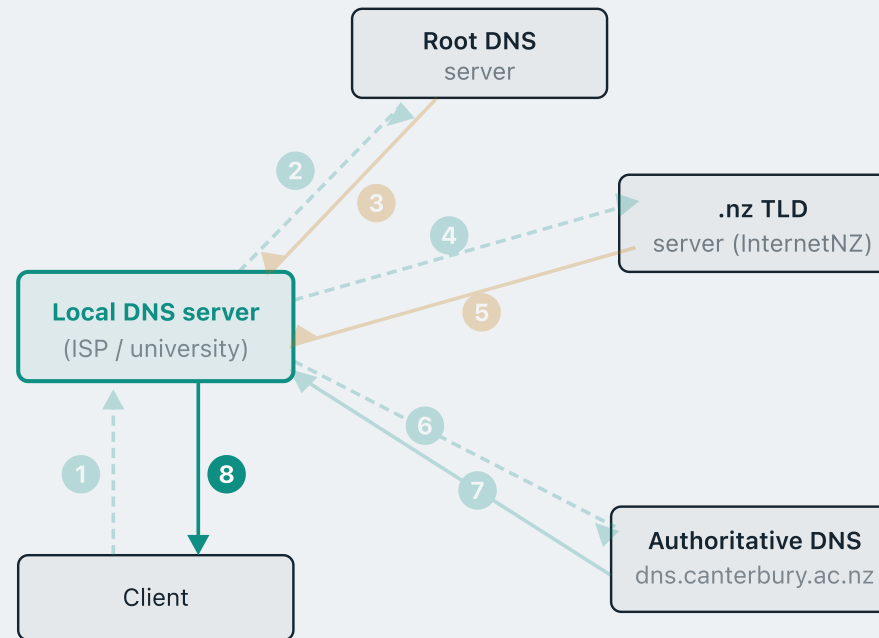
# Resolving a hostname: the full journey

**Step 7.** The authoritative server has the answer! It replies: "csse.canterbury.ac.nz = 132.181.2.21"



# Resolving a hostname: the full journey

**Step 8.** The local DNS returns **132.181.2.21** to the client — and **caches** the mapping for future queries.



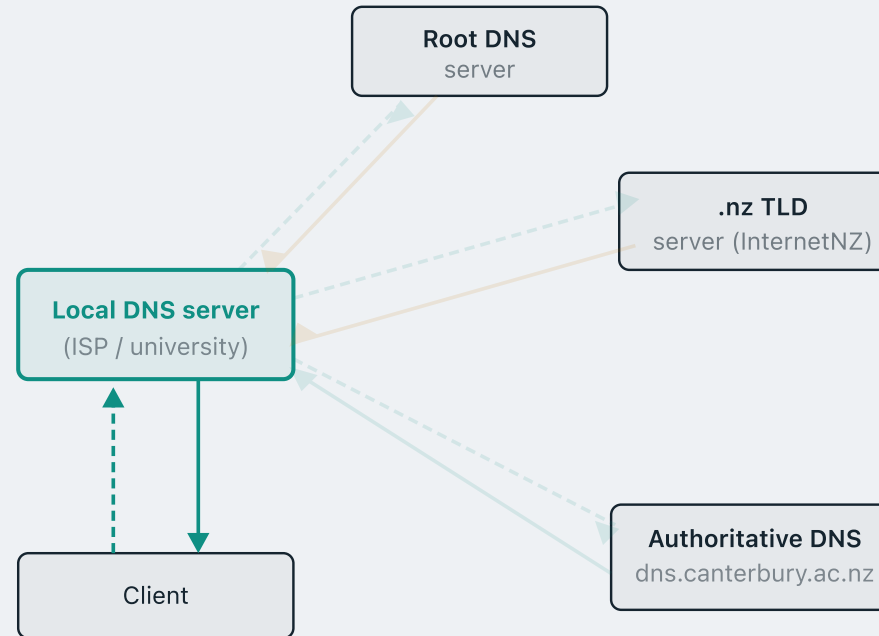
# Recursive vs iterative queries

## Iterative query

The contacted server does **not** resolve the name for you. It returns a **referral** — "I don't know, try that server."

## Recursive query

The contacted server takes **full responsibility** — it resolves the name itself (querying the next server in the chain) and returns the **final answer**.



# Recursive vs iterative queries

## Iterative query

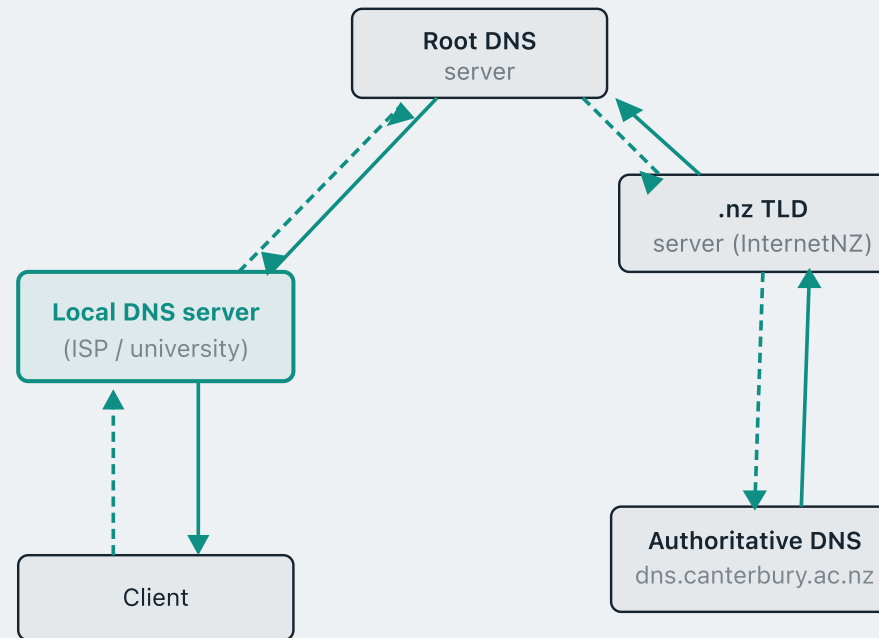
The contacted server does **not** resolve the name for you. It returns a **referral** — "I don't know, try that server."

## Recursive query

The contacted server takes **full responsibility** — it resolves the name itself (querying the next server in the chain) and returns the **final answer**.

## Can we go fully recursive?

In principle, yes — root asks TLD, which asks auth. In practice, root and TLD servers refuse recursion (too much load), so only client → local DNS is recursive.



# DNS Resource Records

DNS distributes data as **Resource Records (RRs)**. Each record has four fields:

(name, value, type, TTL – Time-To-Live)

| NAME                   | VALUE                  | TYPE  | TTL  |
|------------------------|------------------------|-------|------|
| web.canterbury.ac.nz   | 132.181.106.9          | A     | XXXX |
| itdns.canterbury.ac.nz | 132.181.2.225          | A     | XXXX |
| mail.canterbury.ac.nz  | 132.181.3.225          | A     | XXXX |
| canterbury.ac.nz       | itdns.canterbury.ac.nz | NS    | XXXX |
| www.canterbury.ac.nz   | web.canterbury.ac.nz   | CNAME | XXXX |
| canterbury.ac.nz       | mail.canterbury.ac.nz  | MX    | XXXX |

# DNS Resource Records

Type A

**name** = hostname · **value** = IPv4 address

| NAME                   | VALUE                  | TYPE  | TTL  |
|------------------------|------------------------|-------|------|
| web.canterbury.ac.nz   | 132.181.106.9          | A     | XXXX |
| itdns.canterbury.ac.nz | 132.181.2.225          | A     | XXXX |
| mail.canterbury.ac.nz  | 132.181.3.225          | A     | XXXX |
| canterbury.ac.nz       | itdns.canterbury.ac.nz | NS    | XXXX |
| www.canterbury.ac.nz   | web.canterbury.ac.nz   | CNAME | XXXX |
| canterbury.ac.nz       | mail.canterbury.ac.nz  | MX    | XXXX |

# DNS Resource Records

Type NS

**name** = domain · **value** = hostname of the authoritative DNS server for that domain

*"I don't know — but here's the name of the server that does."*

| NAME                   | VALUE                  | TYPE  | TTL  |
|------------------------|------------------------|-------|------|
| web.canterbury.ac.nz   | 132.181.106.9          | A     | XXXX |
| itdns.canterbury.ac.nz | 132.181.2.225          | A     | XXXX |
| mail.canterbury.ac.nz  | 132.181.3.225          | A     | XXXX |
| canterbury.ac.nz       | itdns.canterbury.ac.nz | NS    | XXXX |
| www.canterbury.ac.nz   | web.canterbury.ac.nz   | CNAME | XXXX |
| canterbury.ac.nz       | mail.canterbury.ac.nz  | MX    | XXXX |

# DNS Resource Records

Type CNAME

**name** = alias hostname · **value** = canonical (real) hostname it maps to

| NAME                   | VALUE                  | TYPE  | TTL  |
|------------------------|------------------------|-------|------|
| web.canterbury.ac.nz   | 132.181.106.9          | A     | XXXX |
| itdns.canterbury.ac.nz | 132.181.2.225          | A     | XXXX |
| mail.canterbury.ac.nz  | 132.181.3.225          | A     | XXXX |
| canterbury.ac.nz       | itdns.canterbury.ac.nz | NS    | XXXX |
| www.canterbury.ac.nz   | web.canterbury.ac.nz   | CNAME | XXXX |
| canterbury.ac.nz       | mail.canterbury.ac.nz  | MX    | XXXX |

# DNS Resource Records

Type MX

**name** = domain · **value** = hostname of the mail server for that domain

| NAME                   | VALUE                  | TYPE  | TTL  |
|------------------------|------------------------|-------|------|
| web.canterbury.ac.nz   | 132.181.106.9          | A     | XXXX |
| itdns.canterbury.ac.nz | 132.181.2.225          | A     | XXXX |
| mail.canterbury.ac.nz  | 132.181.3.225          | A     | XXXX |
| canterbury.ac.nz       | itdns.canterbury.ac.nz | NS    | XXXX |
| www.canterbury.ac.nz   | web.canterbury.ac.nz   | CNAME | XXXX |
| canterbury.ac.nz       | mail.canterbury.ac.nz  | MX    | XXXX |

# DNS protocol message format

DNS uses the **same message format** for both **queries** and **replies** — a fixed 12-byte header followed by four variable-length sections.

| identification                                                | flags                    | ← 12 bytes |
|---------------------------------------------------------------|--------------------------|------------|
| number of questions                                           | number of answer RRs     |            |
| number of authority RRs                                       | number of additional RRs |            |
| questions<br>variable number of questions                     |                          |            |
| answers<br>variable number of resource records                |                          |            |
| authority<br>variable number of resource records              |                          |            |
| additional information<br>variable number of resource records |                          |            |

# DNS protocol message format

**Identification**

16-bit number assigned by the client. The reply copies this same value, so the client can match each reply to its original query.

| identification                                                | flags                    | ← 12 bytes |
|---------------------------------------------------------------|--------------------------|------------|
| number of questions                                           | number of answer RRs     |            |
| number of authority RRs                                       | number of additional RRs |            |
| questions<br>variable number of questions                     |                          |            |
| answers<br>variable number of resource records                |                          |            |
| authority<br>variable number of resource records              |                          |            |
| additional information<br>variable number of resource records |                          |            |

# DNS protocol message format

**Flags**

- Query (0) or reply (1)
- Recursion desired — client requests full recursive resolution
- Recursion available — server indicates it supports recursion

| identification                                                | flags                    | ← 12 bytes |
|---------------------------------------------------------------|--------------------------|------------|
| number of questions                                           | number of answer RRs     |            |
| number of authority RRs                                       | number of additional RRs |            |
| questions<br>variable number of questions                     |                          |            |
| answers<br>variable number of resource records                |                          |            |
| authority<br>variable number of resource records              |                          |            |
| additional information<br>variable number of resource records |                          |            |

# DNS protocol message format

**Count fields**

Four 16-bit counters — one per section below — telling how many records appear in each variable-length section.

| identification                                                | flags                    | ← 12 bytes |
|---------------------------------------------------------------|--------------------------|------------|
| number of questions                                           | number of answer RRs     |            |
| number of authority RRs                                       | number of additional RRs |            |
| questions<br>variable number of questions                     |                          |            |
| answers<br>variable number of resource records                |                          |            |
| authority<br>variable number of resource records              |                          |            |
| additional information<br>variable number of resource records |                          |            |

# DNS protocol message format

Questions

The name and record type being queried. e.g. this  
Type A question asks for the IP address of the domain  
facebook.com.

(facebook.com, A)

| identification                                                | flags                    | ← 12 bytes |
|---------------------------------------------------------------|--------------------------|------------|
| number of questions                                           | number of answer RRs     |            |
| number of authority RRs                                       | number of additional RRs |            |
| questions<br>variable number of questions                     |                          |            |
| answers<br>variable number of resource records                |                          |            |
| authority<br>variable number of resource records              |                          |            |
| additional information<br>variable number of resource records |                          |            |

# DNS protocol message format

**Answers**

Resource Records (RRs) that directly answer the query.

( facebook.com, 157.240.22.35, A, TTL=300 )

| identification                                                | flags                    | ← 12 bytes |
|---------------------------------------------------------------|--------------------------|------------|
| number of questions                                           | number of answer RRs     |            |
| number of authority RRs                                       | number of additional RRs |            |
| questions<br>variable number of questions                     |                          |            |
| answers<br>variable number of resource records                |                          |            |
| authority<br>variable number of resource records              |                          |            |
| additional information<br>variable number of resource records |                          |            |

# DNS protocol message format

**Authority**

NS records pointing to authoritative servers —  
returned in referral responses.

(facebook.com, a.ns.facebook.com, NS)

| identification                                                | flags                    | ← 12 bytes |
|---------------------------------------------------------------|--------------------------|------------|
| number of questions                                           | number of answer RRs     |            |
| number of authority RRs                                       | number of additional RRs |            |
| questions<br>variable number of questions                     |                          |            |
| answers<br>variable number of resource records                |                          |            |
| authority<br>variable number of resource records              |                          |            |
| additional information<br>variable number of resource records |                          |            |

# DNS protocol message format

**Additional**

'Glue' Resource Records (RRs) accompanying the authority section — so you can reach the name server without an extra lookup.

```
(a.ns.facebook.com, 129.134.30.12, A)
```

| identification                                                | flags                    | ← 12 bytes |
|---------------------------------------------------------------|--------------------------|------------|
| number of questions                                           | number of answer RRs     |            |
| number of authority RRs                                       | number of additional RRs |            |
| questions<br>variable number of questions                     |                          |            |
| answers<br>variable number of resource records                |                          |            |
| authority<br>variable number of resource records              |                          |            |
| additional information<br>variable number of resource records |                          |            |

# Query & response: a live example

## QUERY →

Transaction ID 0x6492  
Flags 0x0100 (query)  
Questions 1 Answer RRs 0  
Authority RRs 0 Additional RRs 0

### QUESTIONS

www.facebook.com A

## ← RESPONSE

Transaction ID 0x6492  
Flags 0x8180 (response, no error)  
Questions 1 Answer RRs 2  
Authority RRs 4 Additional RRs 8

### QUESTIONS

www.facebook.com A

### ANSWERS

www.facebook.com CNAME  
star-mini.c10r.facebook.com  
star-mini.c10r.facebook.com A  
31.13.78.35

### AUTHORITY

c10r.facebook.com NS  
d.ns.c10r.facebook.com (+3)

### ADDITIONAL

d.ns.c10r.facebook.com A  
185.89.219.11 (+7)

# DNS — key takeaways

- **Main purpose** — DNS maps human-readable hostnames to IP addresses.
- **Hierarchy** — DNS is distributed: no single server holds all records.  
Responsibility is delegated across Root, TLD, and Authoritative servers.
- **Traversal pattern** — the client queries the local DNS resolver, which then follows referrals Root → TLD → Authoritative.  
Results are cached locally by TTL to avoid repeated lookups.

## References

James F. Kurose & Keith W. Ross, *Computer Networking: A Top-Down Approach*, Addison Wesley, 3<sup>rd</sup> edition, 2005.

Chapter 2 Application Layer (2.1, 2.2, 2.4, 2.5)

William Stallings, *Data and Computer Communications*, Pearson, 10<sup>th</sup> edition, 2013.

Chapter 24 Electronic Mail, DNS, and HTTP (24.1–24.3)

Andrew S. Tanenbaum & David J. Wetherall, *Computer Networks*, Prentice Hall, 5<sup>th</sup> edition, 2010.

Chapter 7 The Application Layer (7.1–7.3)

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Prof Aleksandar Kuzmanovic's lecture notes for CS340, Northwestern University, 2005